Graphs & Transformations

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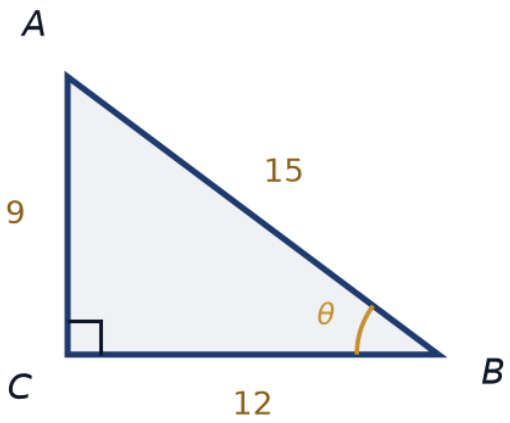
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On an Argand diagram, the point P represents the complex number z such that $|z - (2 + 3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

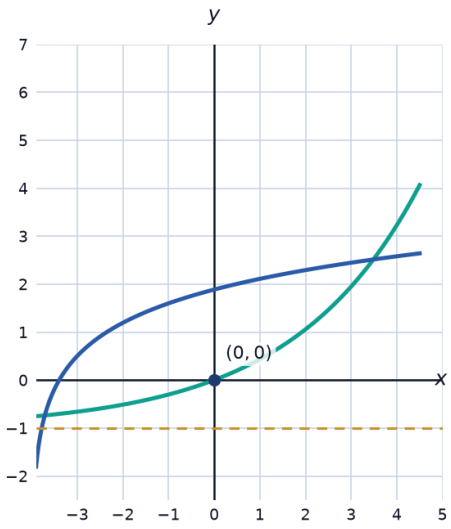
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Practice 1.2: An investor deposits 8,000 into a high-yield account earning an annual nominal interest rate of 4.8% compounded monthly. Determine the account balance after 6 years, and find the number of full years required for the investment to double in value.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the logarithmic equation $\log_2(x+3) + \log_2(x-3) = 4$ for all real solutions x. State any extraneous roots.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 17.01 Discrete Random Variables

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 18.01 Binomial Distribution

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 19.01 Normal Distribution

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

Lines ℓ_1 and ℓ_2 have direction vectors $\mathbf{d_1} = (2; 1; -1)$ and $\mathbf{d_2} = (1; -1; 3)$. Find the acute angle between them.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Trigonometry

An IB AA SL student scores 64/80 on Paper 1 (40%), 72/80 on Paper 2 (40%), and 16/20 on the Internal Assessment (20%). Calculate the student's overall weighted percentage.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Describe the transformation that maps the curve $y = f(x)$ onto $y = f(x - 3) + 5$.

a. Translation by vector $[3, 5]$

b. Translation by vector $[-3, 5]$

c. Translation by vector $[3, -5]$

d. Translation by vector $[-3, -5]$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve $y = x^2$ is translated by vector $[-2, -4]$. What is the equation of the resulting curve?

a. $y = (x + 2)^2 - 4$

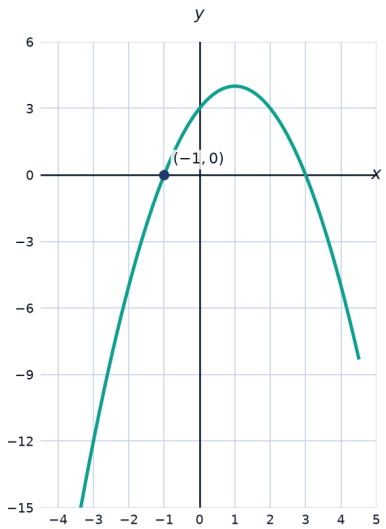
b. $y = (x - 2)^2 - 4$

c. $y = (x + 2)^2 + 4$

d. $y = x^2 - 2x - 4$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coordinates of the turning point of $y = 4 - (x - 1)^2$.



a. $(1, 4)$

b. $(-1, 4)$

c. $(1, -4)$

d. $(4, 1)$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Describe the transformation that maps $y = f(x)$ onto $y = -f(2x)$.

a. Reflection in x-axis followed by horizontal stretch scale factor $1/2$

b. Reflection in y-axis followed by vertical stretch scale factor 2

c. Reflection in x-axis followed by horizontal stretch scale factor 2

d. Reflection in y-axis followed by horizontal stretch scale factor $1/2$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The point $(4, 12)$ lies on the curve $y = f(x)$. What are the coordinates of the corresponding point on $y = 3f(x/2)$?

a. $(8, 36)$

b. $(2, 36)$

c. $(8, 4)$

d. $(2, 4)$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which equation represents the reflection of the curve $y = 2^x$ in the y-axis?

a. $y = 2^{-x}$

b. $y = -2^x$

c. $y = -2^{-x}$

d. $y = \log_2(x)$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = x^3$ undergoes a stretch scale factor 2 parallel to the y-axis, followed by a translation by vector $[1, -3]$. Find its equation.

a. $y = 2(x - 1)^3 - 3$

b. $y = (2x - 1)^3 - 3$

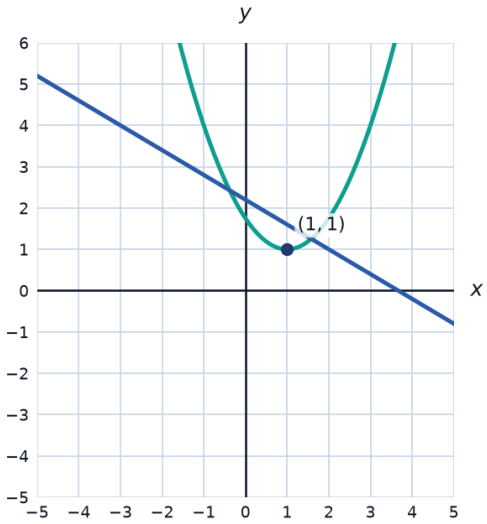
c. $y = 2(x + 1)^3 - 3$

d. $y = 2x^3 - 5$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A curve with equation $y = g(x)$ has a local minimum at $(3, -5)$. Where is the corresponding local extremum on the curve $y = -2g(x + 1)$?



a. Local maximum at $(2, 10)$

b. Local minimum at $(4, 10)$

c. Local maximum at $(4, -10)$

d. Local minimum at $(2, -10)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the asymptotes of the rational function $y = \frac{2x + 1}{x - 3}$.

a. $x = 3$ and $y = 2$

b. $x = -3$ and $y = 2$

c. $x = 3$ and $y = 1$

d. $x = 1/2$ and $y = 3$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the coordinates of all intersection points of $y = (x - 2)^2(x + 1)$ with the coordinate axes.

a. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, 4)$

b. x-intercepts at $(-2, 0)$ and $(1, 0)$; y-intercept at $(0, -4)$

c. x-intercept at $(2, 0)$; y-intercept at $(0, 4)$

d. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, -2)$

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The curve $y = \frac{a}{x+b}$ passes through $(1, 6)$ and $(4, 2)$. Find the values of a and b.

a. $a = 12, b = 1$

b. $a = 6, b = 0$

c. $a = 18, b = 2$

d. $a = 8, b = 0$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Determine the number of real solutions to the equation $|x^2 - 4| = 3$.

a. 4 solutions

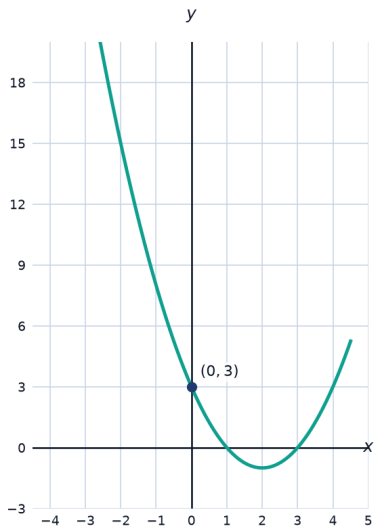
b. 2 solutions

c. 3 solutions

d. 0 solutions

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.



a. (4, 3)

b. (3, 2)

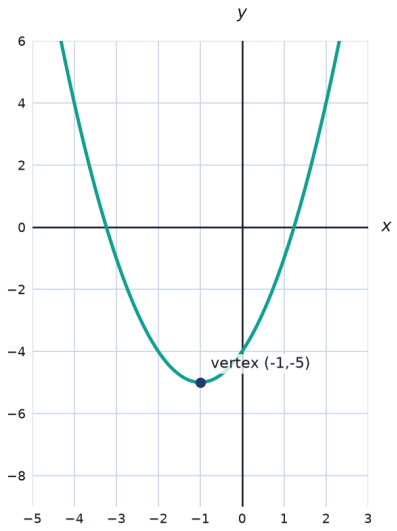
c. (5, 4)

d. (2, 1)

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 5 units down

b. Shifting 1 units right and 5 units down

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the functions $f(x) = |x - a|$ and $g(x) = b$, where a and b are constants. The graph of $y = f(x)$ is a V-shaped graph with its vertex at $(a, 0)$. The graph of $y = g(x)$ is a horizontal line. If the graphs of $y = f(x)$ and $y = g(x)$ intersect at exactly one point, what is the relationship between a and b for the graphs to intersect at a single point?

a. a = b

b. a = -b

c. b = 0

d. a = 0

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Convert 210° to radians, giving your answer as an exact multiple of π.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$

d. $\frac{7\pi}{4}$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k is $f(x) = -2x + k$ (for $x < 5$) and $f(x) = x^2$ (for $x \geq 5$) continuous at $x = 5$?

a. 36

b. 34

c. -35

d. 35

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A root of $f(x)$ lies in $[-6, -3]$, where $f(-6)$ is negative and $f(-3)$ is positive. After one bisection step (midpoint $m = \frac{-6-3}{2} = -\frac{9}{2}$), $f(m)$ is negative. Which sub-interval contains the root?

a. $[-6, -\frac{9}{2}]$

b. $[-\frac{9}{2}, -3]$

c. [-6,-3]

d. [-7,-6]

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

Given $f(x) = (x-2)^2(x+1)$, does the graph touch or cross the x-axis at $x = 2$?

a. The graph does not intersect the x-axis there

b. Not enough information

c. Touches (does not cross)

d. Crosses

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Trigonometry

If $f(x) = \sqrt{x-5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \leq x \leq 3$

b. $x \leq -3$ or $x \geq 3$

c. $x \geq 3$

d. All real numbers

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Differentiation & Integration

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. -5

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Statistics & Probability

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

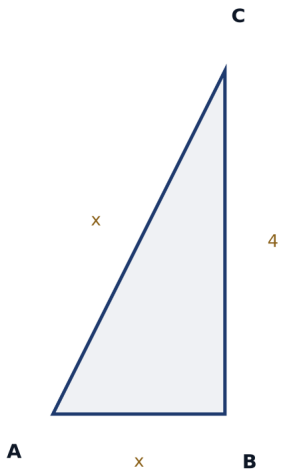
b. Odd

c. Neither

d. Cannot be determined

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Functions: Composite & Inverse

In the diagram, triangle ABC is shown with coordinates A(2, 1), B(7, 1), and C(5, 5). What is the length of the side AC?



a. 4

b. 5

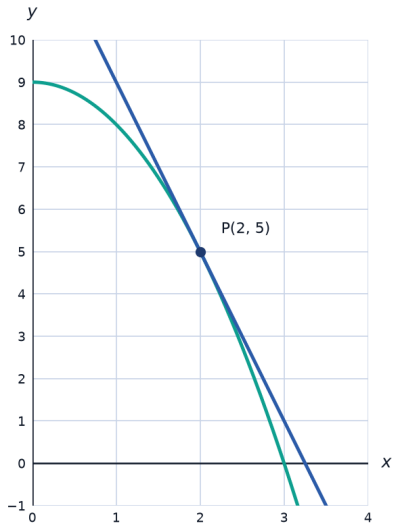
c. $2\sqrt{5}$

d. $\sqrt{20}$

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44 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 9.01 Circular Measure**

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.



a. $\frac{11}{24}$

b. $\frac{125}{192}$

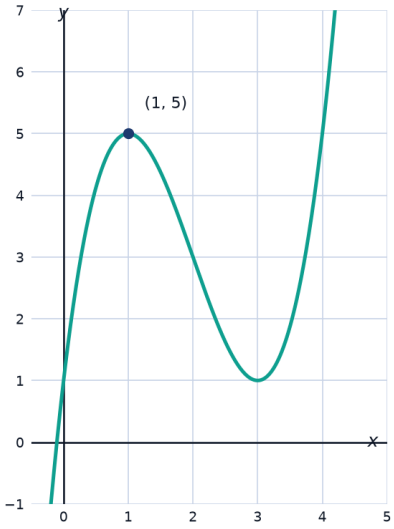
c. $\frac{8}{3}$

d. $\frac{25}{8}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$. Find the coordinates of the other stationary point on the curve.



a. $(3, 1)$

b. $(3, 0)$

c. $(3, -1)$

d. $(-3, 1)$

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